MIRA AI Triage Platform

Medical Intelligent Response Assistant

AI-powered platform designed to help hospitals classify and prioritize patients efficiently using configurable triage rules and intelligent decision systems.

Real-time Analysis

Instant patient vitals processing

Smart Classification

AI-driven severity assessment

Efficient Triage

Optimized resource allocation

The Challenge

Hospitals face major operational challenges

Overcrowded EDs

Emergency departments overwhelmed with patient volume

Staff Shortages

Insufficient medical professionals available

Delayed Identification

Critical patients not identified quickly enough

Inefficient Triage

Manual processes lead to poor resource allocation

These issues lead to slower treatment and poor resource allocation

Solution

MIRA AI Triage Platform

AI-assisted healthcare solution for intelligent patient management

Automatic Vitals Analysis

Real-time patient vital sign processing

Severity Classification

Critical | Moderate | Stable

Resource Allocation

Intelligent assignment of doctors and wards

Process Flow

Patient Vitals
Data Input

→

AI Evaluation
Intelligent Analysis

→

Severity Classification
Priority Level

→

Resource Allocation
Assignment

Automated workflow ensures faster and more accurate patient prioritization

Operational Workflow

1

Administrator Defines Rules

Admin configures triage classification logic using AI-powered rule engine

Configuration Phase

2

Patient Vitals Entered

Manager inputs patient vital signs and symptoms into the system

Data Input

3

AI Evaluates Condition

AI models analyze patient data and apply triage rules

AI Processing

4

Patient Categorized

System assigns severity level: Critical, Moderate, or Stable

Critical | Moderate | Stable

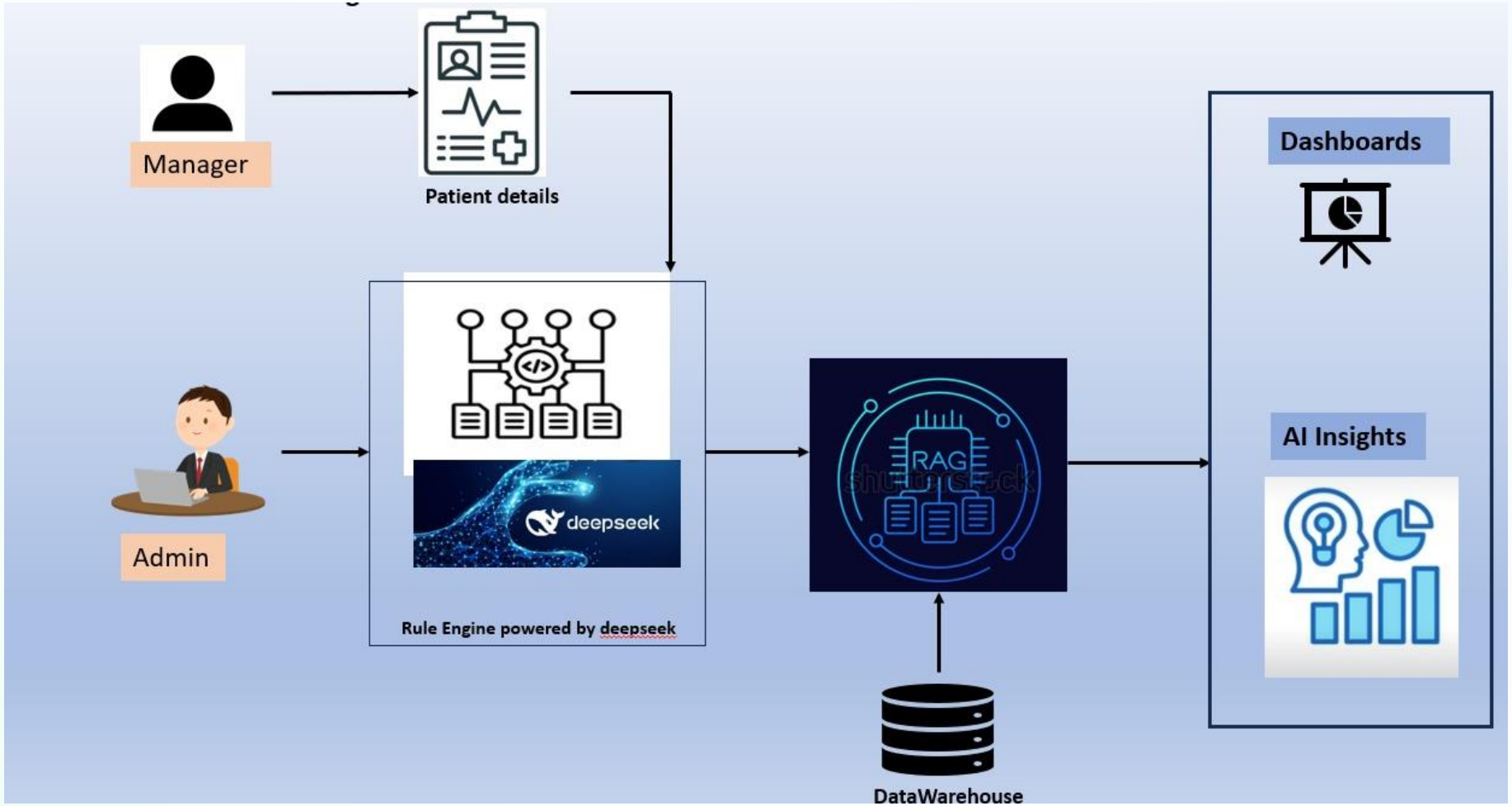
5

Dashboard Display

Manager views prioritized patient list with actionable insights

Action Phase

Architecture



Technology Stack

Frontend

React
Modern component-based UI library

Tailwind CSS
Utility-first styling framework

Framer Motion
Smooth animations & transitions

Recharts
Interactive data visualization

Backend

FastAPI
High-performance Python framework

SQLAlchemy
ORM for database operations

Pydantic
Data validation & settings

SQLite
Lightweight embedded database

AI Layer

GPT-4o
Advanced reasoning & explanations

GPT-4o Mini
Fast inference & lightweight

DeepSeek R1
Structured reasoning engine

DeepSeek V3
Analytical reasoning & insights

ER Diagram

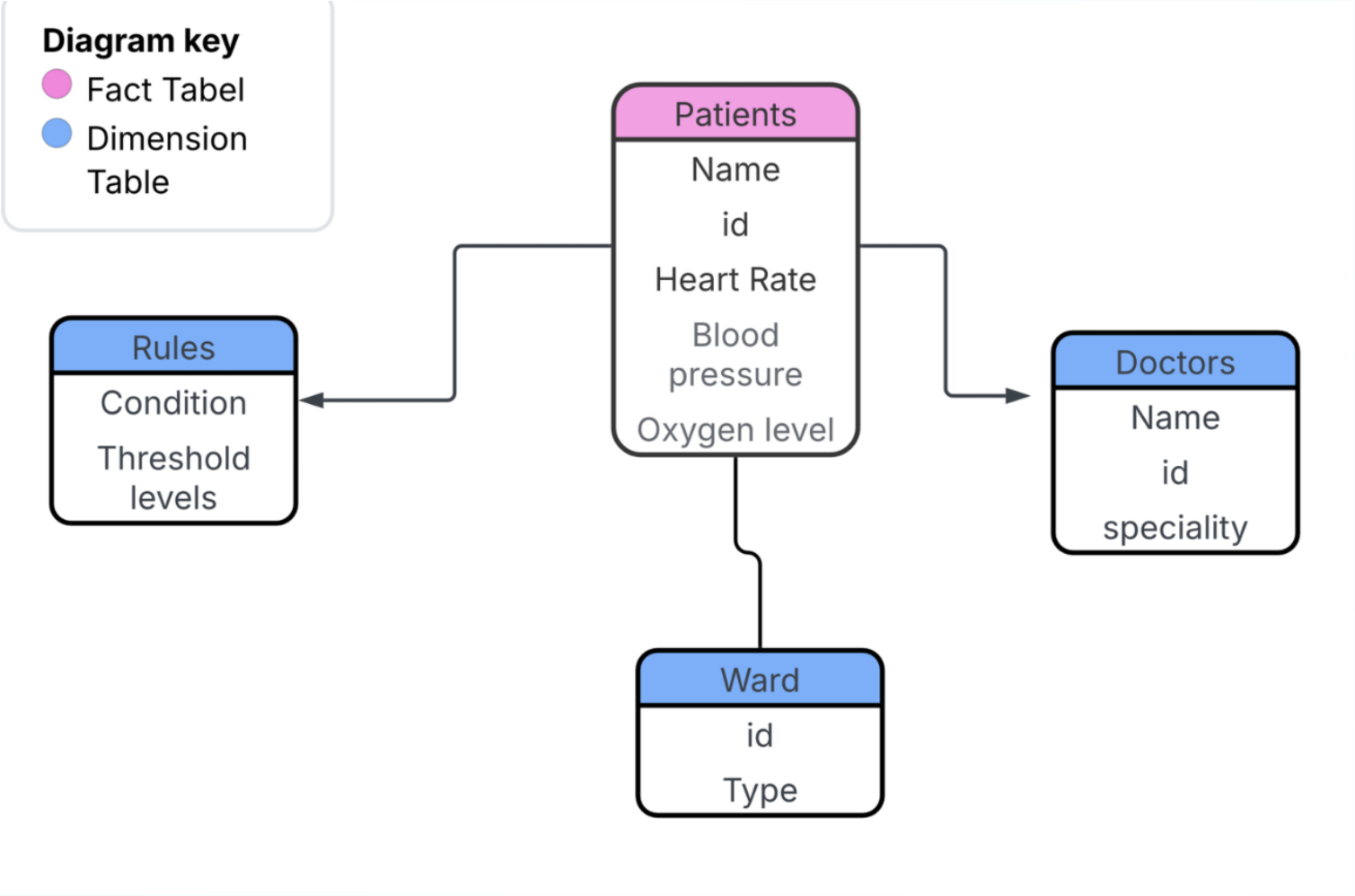


Diagram key
Fact Tabel
Dimension Table
Patients
Name
id
Heart Rate
Blood pressure
Oxygen level
Rules
Condition
Threshold levels
Doctors
Name
id
speciality
Ward
id
Type

AI Models Integration

GPT-4o

Advanced Reasoning

- Natural language reasoning
- Prompt interpretation
- Structured output generation

GPT-4o Mini

Fast & Efficient

- Lightweight inference
- Quick response time
- Real-time dashboard insights

DeepSeek R1

Structured Logic

- Rule-based reasoning
- Decision logic evaluation
- Symptom-to-ward mapping

DeepSeek V3

Advanced Analysis

- Analytical reasoning
- Multi-step logic processing
- Detailed clinical insights

Multi-Model Strategy

Different tasks require different AI strengths — reasoning, speed, or structured analysis.

Accuracy

Specialized models for each task

Performance

Optimized response times

Scalability

Flexible architecture design

Why & Where: Model Applications

DeepSeek V3

Generate AI Insights

Purpose:

Patient condition explanation & medical reasoning

Example Use Case:

"Patient classified as Critical due to low oxygen levels and elevated heart rate indicating respiratory distress."

GPT-4o

Create Rule from Prompt

Purpose:

Convert natural language into structured triage rules

Example Use Case:

"If oxygen level drops below 88 assign ICU" → Condition: oxygen < 88, Severity: Critical, Ward: ICU

DeepSeek R1

Assign Ward Based on Symptoms

Purpose:

Perform structured decision logic for patient placement

Example Use Case:

Symptoms: Chest pain, Low oxygen, High heart rate → Severity: Critical, Ward: ICU, Doctor: Cardiologist

GPT-4o Mini

Patient Insight in Monitor

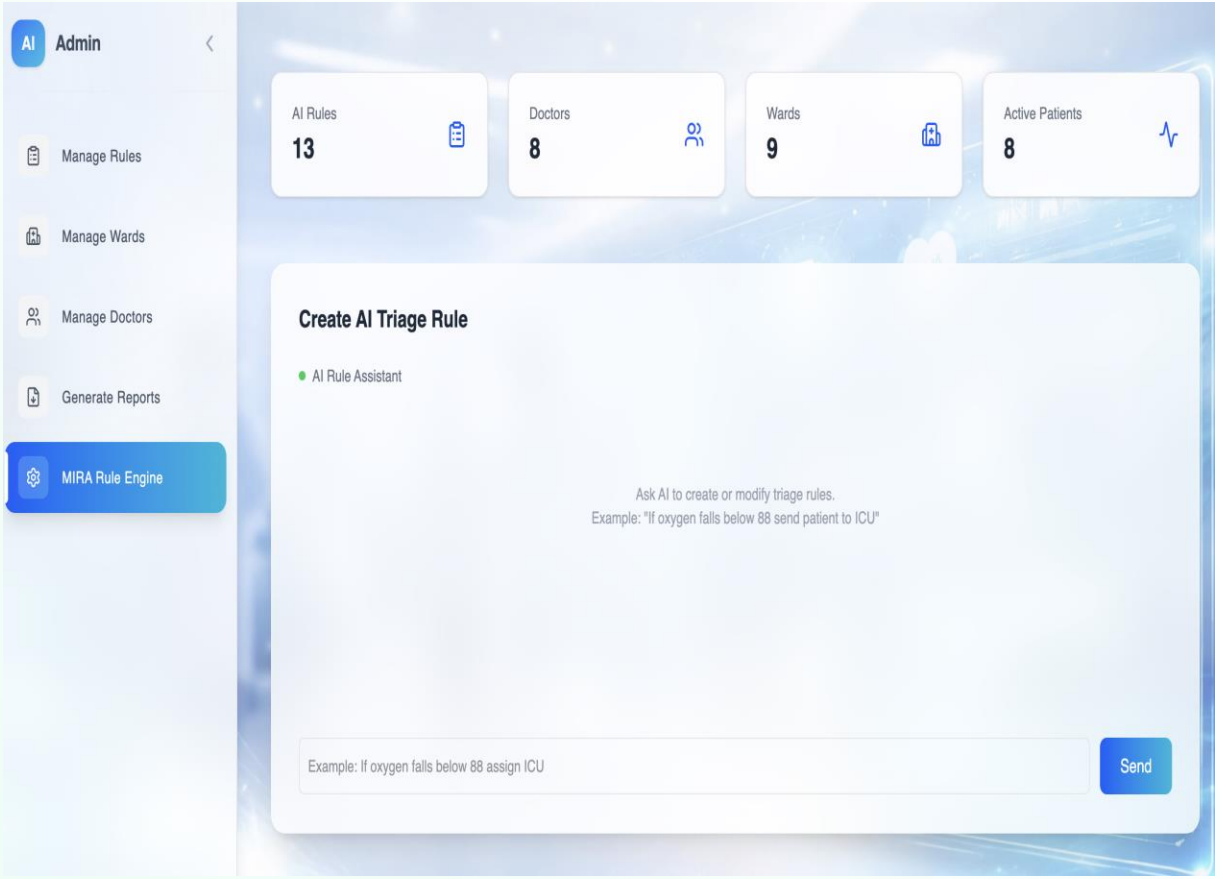
Purpose:

Quick AI explanations in Patient Monitor dashboard

Example Use Case:

"Patient classified as Moderate because oxygen levels are stable but temperature is elevated indicating infection risk."

Administrator Portal



Admin Capabilities

Configure Triage Rules

AI-powered rule engine for intelligent classification logic

Manage Hospital Wards

Oversee ward assignments and resource allocation

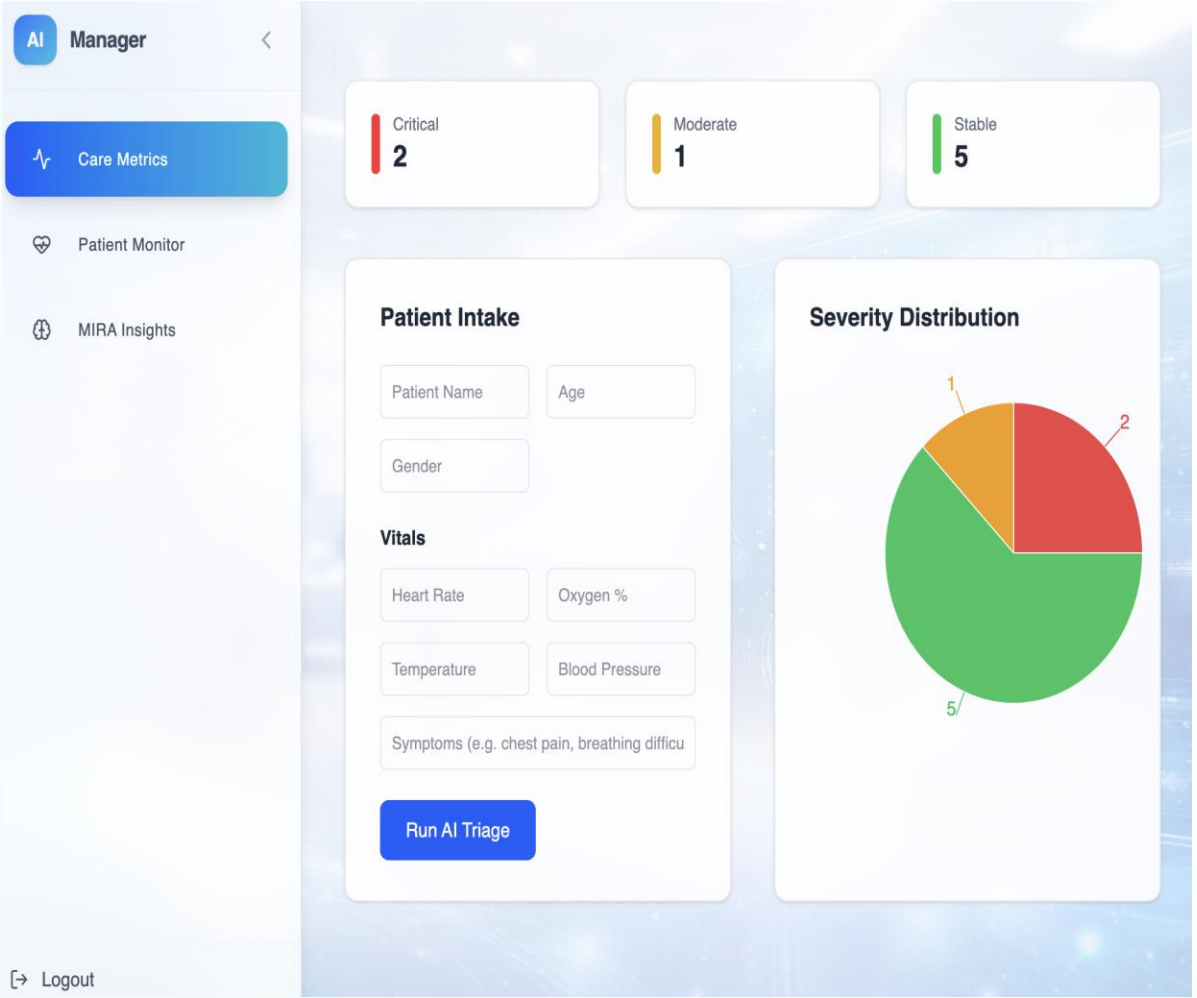
Manage Doctors

Coordinate medical staff assignments and specializations

Export System Reports

Generate comprehensive analytics and operational data

Manager Dashboard



Dashboard Features

Patient Intake Form

Enter patient vitals and symptoms for AI analysis

Triage Classification Results

AI-powered severity assessment and prioritization

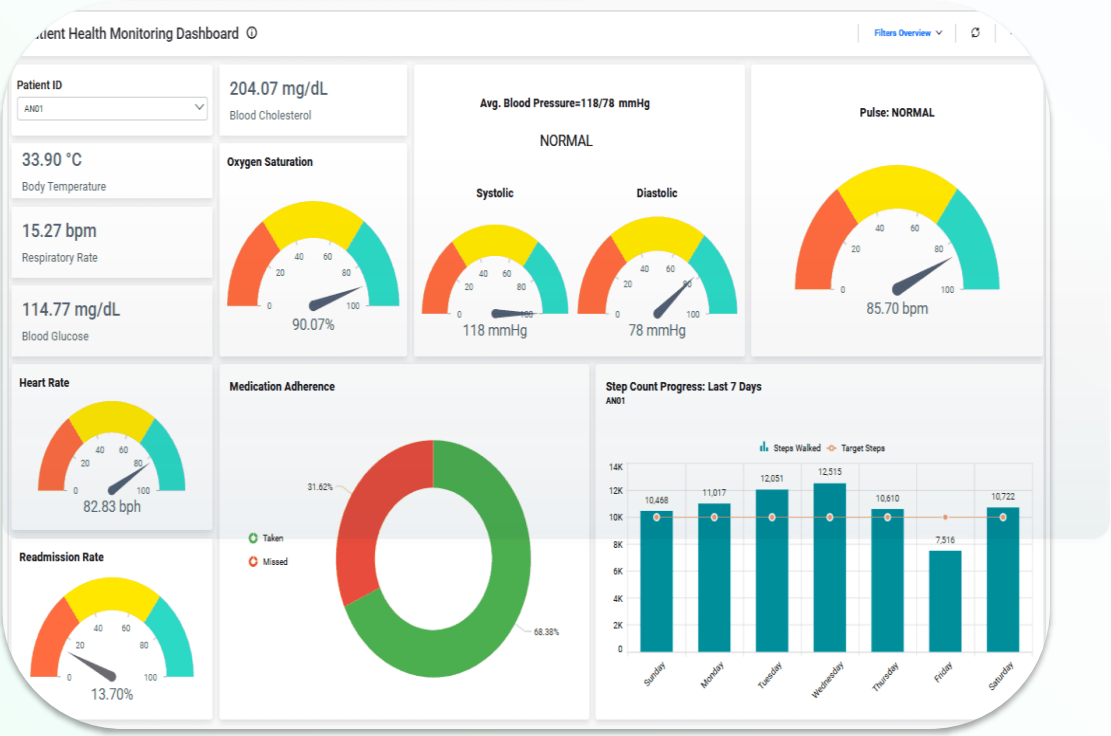
Severity Distribution

Visual overview of patient priority levels

Real-time Monitoring

Track patient status and critical alerts

Key Benefits



Faster Patient Prioritization

AI-powered triage reduces decision time by 60%

Improved Resource Allocation

Intelligent assignment of doctors, wards, and equipment

Reduced Treatment Delays

Critical patients identified and treated faster

AI-Powered Insights

Clinical decision support for healthcare teams

Future Roadmap

EHR Integration

Electronic Health Records

Connect with existing hospital EHR systems for seamless patient data synchronization and comprehensive medical history access.

→ Phase 1 Implementation

Predictive Risk Analysis

Proactive Patient Management

Identify patient risks before critical events occur using advanced AI predictive models and early warning indicators.

→ AI Enhancement

AI Clinical Support

Treatment Recommendations

AI-powered clinical decision support providing treatment suggestions, drug interaction checks, and clinical alerts.

→ Clinical Intelligence

Real-time ICU Monitoring

Continuous Vital Signs

IoT-enabled continuous monitoring of ICU patients with real-time vital signs streaming and automated alerts.

→ IoT Integration

Live Demo

See MIRA in Action

AI Triage

Real-time Dashboard

Instant Classification

Experience real-time AI-powered patient classification and intelligent resource allocation

Thank You

Questions & Discussion

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Transforming Healthcare with AI-Powered Intelligence